Practice workbook · Calculating percentages

Percentage of a total, percentage of a quantity and percentage change.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-12

1. Question 1 · part c · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c only.

1 Thomas recruited participants to investigate obedience to authority. The participants were asked to attend a university.

They were allocated to one of two conditions:

- Condition A: 30 participants were instructed to answer general knowledge questions in silence for 40 minutes by a man dressed in a uniform.
- Condition B: 30 participants were instructed to answer general knowledge questions in silence for 40 minutes by a man dressed in jeans and a t-shirt.

Thomas recorded how many participants obeyed the instruction to remain silent.

(a) Identify the dependent variable (DV) in Thomas' investigation.

(1)

- ☒ **A** Whether the man was wearing a uniform or jeans and a t-shirt.
- ☒ **B** The amount of time participants were given to answer the questions.
- ☒ **C** The number of participants who obeyed the man's instruction to remain silent.
- ☒ **D** The number of general knowledge questions participants were instructed to answer.

(b) Describe **one** participant variable that Thomas may have considered when conducting his investigation.

(2)

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1. Question 1 · part c · 2 marks

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Worked-example retry · Attempt part c only.

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The results of Thomas' investigation are shown in **Table 1**.

	Obeyed instruction (remained silent)	Disobeyed instruction (did not remain silent)
Condition A: Man wearing uniform	22	8
Condition B: Man wearing jeans and t-shirt	11	19

Table 1

(c) Calculate the percentage of participants who **obeyed** the instruction, out of all the participants in the investigation.

(2)

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Percentage .....

(d) Calculate the ratio of participants who **disobeyed** the instruction, to all participants taking part in the investigation.

(2)

Ratio .....

(e) Thomas concludes that the clothes worn by an authority figure influence levels of obedience.

Give **one** way Thomas' results could support this conclusion.

(1)

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(Total for Question 1 = 8 marks)

1. Question 1 · part c · 2 marks · Working and answer

Working and answer

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2. Question 1 · part d.i · 2 marks

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Further practice · Attempt part d.i only.

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1 Matthew investigated the effectiveness of drug treatment for patients with depression. He interviewed six patients from a local mental health service provider before and after drug treatment.

Matthew only used preset questions in his interview.

(a) Identify the type of interview Matthew used in his investigation.

(1)

☒ **A** Structured interview

☒ **B** Unstructured interview

☒ **C** Semi-structured interview

☒ **D** Covert interview

(b) Matthew made sure that the participants in his interviews had the right to withdraw.

Define what is meant by the 'right to withdraw'.

(1)

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(c) Give **two** ways that Matthew could ensure the confidentiality of the patients he interviewed.

(2)

1 .....

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2 .....

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P 5 9 6 6 9 A 0 3 4 8

2. Question 1 · part d.i · 2 marks

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Further practice · Attempt part d.i only.

(d) Participant responses to the question 'How often do you feel tired?' were scored using a scale of 0 to 5, with 0 being 'never feel tired' and 5 being 'always feel tired'.

The results of Matthew's investigation are shown in **Table 1**.

Participant	Score (out of 5) for tiredness before drug treatment	Score (out of 5) for tiredness after drug treatment
A	5	3
B	4	3
C	5	2
D	3	0
E	5	1
F	4	1

Table 1

(i) The score for tiredness of **Participant F** decreased from before drug treatment to after drug treatment.

Calculate the percentage decrease in the score for tiredness of **Participant F**.

(2)

Percentage decrease in score for tiredness of **Participant F** .....

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2. Question 1 · part d.i · 2 marks

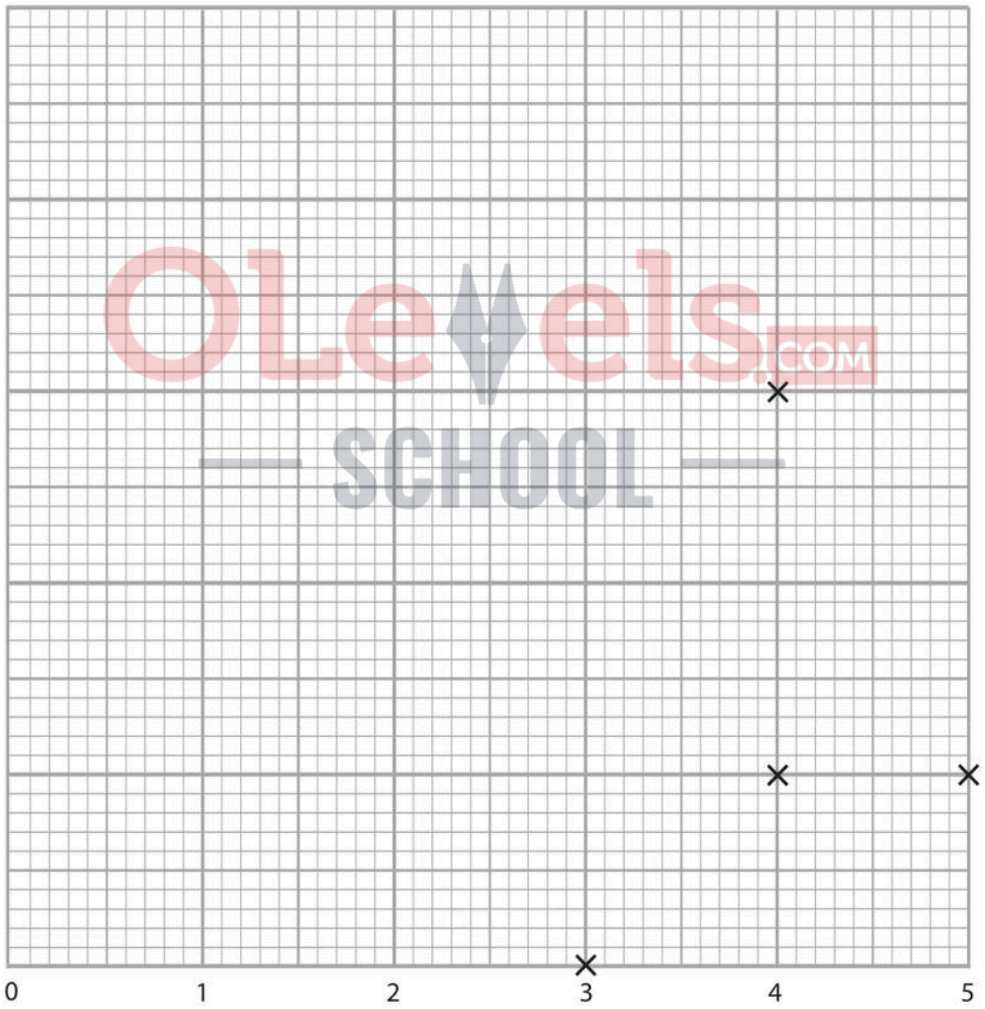
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Further practice · Attempt part d.i only.

(ii) Complete the scatter diagram for **Participants A and C**, using the data in **Table 1**, including the label and scale for the y-axis.

(3)

A scatter diagram to show the relationship between tiredness and drug treatment



Score (out of 5) for tiredness before drug treatment

(Total for Question 1 = 9 marks)

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2. Question 1 · part d.i · 2 marks · Working and answer

Working and answer

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3. Question 6 · part c · 1 mark

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Further practice · Attempt part c only.

6 Kayla investigated whether the use of positive facial expressions (such as smiles) by actors in television advertisements resulted in people being more likely to buy the product being advertised.

First, she analysed two newspaper articles about the use of facial expressions in television advertisements.

Kayla then used an online questionnaire to find out if facial expressions influenced the purchase of a particular brand of cereal. 39,980,000 responses were collected in total.

(a) Give 39,980,000 in standard form.

(1)

39,980,000 in standard form .....

(b) 10,003,010 responses showed positive facial expressions used by actors in television advertisements did not influence their decision to buy the cereal.

Estimate the number of responses that showed positive facial expressions **did** influence their decision to buy the cereal.

(1)

Estimation .....

(c) 63% of the total responses to the online questionnaire were completed by families with children.

Calculate the number of families with children who completed the online questionnaire.

(1)

Number of families with children who completed the online questionnaire .....

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3. Question 6 · part c · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c only.

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(d) Complete **Table 4** to show which data used by Kayla in her investigation was primary data and which was secondary data.

(2)

Type of data	Example from Kayla's investigation
Primary data	
Secondary data	

Table 4

(Total for Question 6 = 5 marks)

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Working and answer

4. Question 4 · part a · 2 marks

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Further practice · Attempt part a only.

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4 Ricky used a field experiment to investigate whether there is a difference between English and Sociology students' obedience to an authority figure.

He arranged for an English teacher and Sociology teacher to give their students an instruction to complete a piece of work. The teachers then left the classroom for 45 minutes.

Ricky recorded the number of students who did and did not complete the work. The results are shown in **Table 3**.

	Number of students who did complete the work	Number of students who did not complete the work
English students	16	8
Sociology students	15	10

Table 3

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(a) Calculate the percentage of students who completed the work, out of all the students in the experiment.

(2)

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Percentage of students who completed the work .....

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Turn over ▶

4. Question 4 · part a · 2 marks

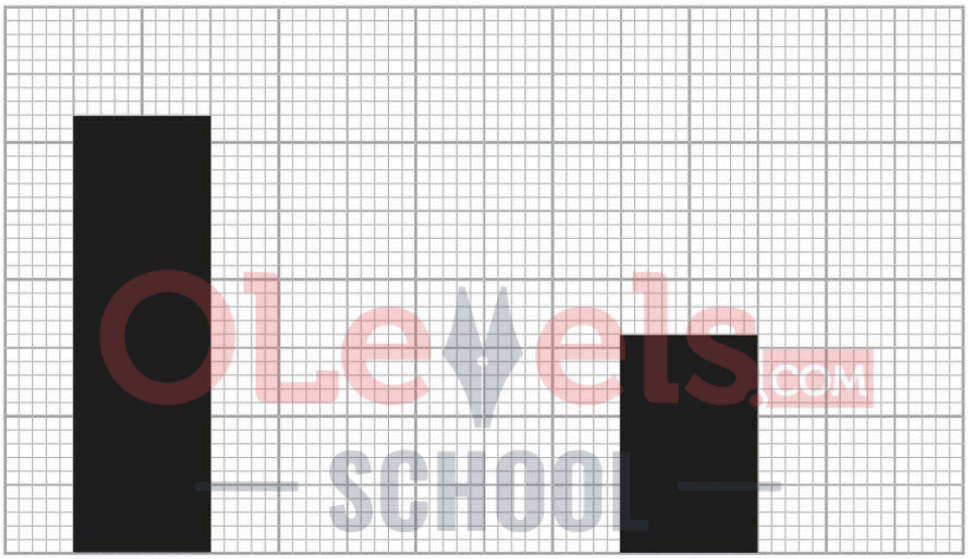
2020 NOV · Paper 02

Further practice · Attempt part a only.

(b) Complete the bar chart with the data from **Table 3** for the Sociology students, including the label and scale for the y-axis.

(3)

A bar chart to show the number of English and Sociology students who did and did not complete the work set by the teacher.



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4. Question 4 · part a · 2 marks · Working and answer

Working and answer

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5. Question 2 · part c.ii · 2 marks

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Further practice · Attempt part c.ii only.

2 Yolanda is conducting an observation of children's play in an early years day care setting. She observed children in the setting for four hours on a Monday.

(a) Define what is meant by an 'observation'.

(1)

(b) Describe how Yolanda could have used an opportunity sampling technique to select the children for her observation.

(2)

(c) Yolanda recorded the number of children who played with the different types of toys in the day care setting.

There were a total of 74 children recorded:

- 23 of the total children played with vehicles
- 11 of the total children played with dolls
- 29 of the total children played with role-play toys.

The remaining children played with arts and crafts toys.

(i) Calculate the ratio of children who played with dolls to those who played with arts and crafts toys.

(2)

Ratio .....

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5. Question 2 · part c.ii · 2 marks

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Further practice · Attempt part c.ii only.

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(ii) Calculate the percentage of children who played with role-play toys as a percentage of the total number of children recorded.

You must give your answer to two decimal places.

(2)

Percentage .....

(d) Yolanda recorded the gender of the children who played with vehicles.

Her results are shown in **Table 1**.

	Number of children who played with vehicles
Males	18
Females	5

Table 1

Explain **one** conclusion that Yolanda could make about children's play from the results in **Table 1**.

(2)

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(Total for Question 2 = 9 marks)

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5. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

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6. Question 6 · part b · 1 mark

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Further practice · Attempt part b only.

6 John researched the effect of a new drug treatment for depression. He had 60 patients with a diagnosis of depression that were split into two conditions.

- Condition A: 20 patients received a placebo sugar pill.
- Condition B: 40 patients received the new drug treatment.

In Condition A, 20% of patients reported a decrease in their symptoms of depression.
In Condition B, 85% of patients reported a decrease in their symptoms of depression.

(a) Calculate how many patients received the placebo sugar pill as a fraction of all patients.

You must give your answer in the lowest form.

(2)

Fraction .....

(b) Calculate the number of patients in **Condition A** who reported a decrease in their symptoms.

(1)

Number of patients .....

(c) John concludes that the new drug treatment is a more effective treatment for depression than a placebo sugar pill.

Give **one** way that John's findings could support this conclusion.

(1)

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(Total for Question 6 = 4 marks)

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6. Question 6 · part b · 1 mark · Working and answer

Working and answer

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7. Question 1 · part d · 2 marks

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Further practice · Attempt part d only.

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1 Andrew conducted a laboratory experiment to investigate the capacity of short-term memory. He showed participants 20 trigrams and asked them to recall as many of the trigrams as they could.

(a) Andrew counted how many trigrams each participant recalled correctly.

Identify the type of data gathered by Andrew.

(1)

- ☒ **A** Qualitative data
- ☒ **B** Subjective data
- ☒ **C** Quantitative data
- ☒ **D** Secondary data

(b) Andrew used a standardised procedure when showing the trigrams to his participants.

Give **two** ways that Andrew could have standardised his procedure.

(2)

1 .....

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2 .....

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7. Question 1 · part d · 2 marks

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Further practice · Attempt part d only.

(c) Andrew sampled a total of 20 participants. A quarter of the participants were male.

Calculate the ratio of males to females in the participant sample used by Andrew in his experiment.

(2)

Ratio .....

(d) The number of trigrams recalled by participants correctly in Andrew's experiment are shown in **Table 1**.

	Number of trigrams recalled correctly				
	Between 1 and 4 trigrams	**Between 5 and 8 trigrams**	**Between 9 and 12 trigrams**	**Between 13 and 16 trigrams**	**Between 17 and 20 trigrams**
Males	0	2	3	0	0
Females	1	5	8	1	0

Table 1

Calculate the percentage of participants who recalled between 9 and 12 trigrams correctly, out of all the participants in the experiment.

(2)

Percentage .....

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7. Question 1 · part d · 2 marks

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Further practice · Attempt part d only.

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(e) Explain **one** conclusion that Andrew could make about short-term memory from the results in **Table 1**.

(2)

(Total for Question 1 = 9 marks)

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Working and answer

8. Question 1 · part c · 2 marks

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Further practice · Attempt part c only.

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1 Gary conducted an experiment to test whether there was a difference in the ability of boys and girls to conserve number. He sampled 50 boys and 50 girls aged six years old.

Gary showed them the image in **Figure 1**.

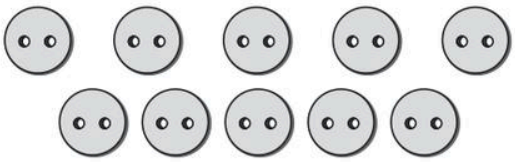


Figure 1

Gary asked each child to say whether the top row of buttons contained more, less, or the same number of buttons as the bottom row.

(a) Identify the independent variable in Gary's experiment.

(1)

- ☒ **A** age
- ☒ **B** gender
- ☒ **C** ability
- ☒ **D** conservation

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(b) Give **two** ways that Gary could ensure his experiment is ethical.

(2)

1 .....

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2 .....

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8. Question 1 · part c · 2 marks

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Further practice · Attempt part c only.

The results of Gary's experiment are shown in **Table 1**.

	Number of children who said the top row contained less buttons than the bottom row	Number of children who said the top row contained the same number of buttons as the bottom row	Number of children who said the top row contained more buttons than the bottom row
Boys	5	13	32
Girls	1	25	24

Table 1

(c) Calculate the percentage of girls who said that the top row contained the **same** number of buttons as the bottom row, out of all the children in the experiment.

(2)

Percentage .....

(d) Calculate the range for the number of children who said the top row contained **more** buttons than the bottom row.

(1)

Range .....

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8. Question 1 · part c · 2 marks · Working and answer

Working and answer

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9. Question 4 · part a · 1 mark

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Further practice · Attempt part a only.

4 Pedro was investigating whether schemas have an influence on memory accuracy. He decided to test four students from his psychology class.

He placed 10 typical objects, such as a chair, and 10 unusual objects, such as a laundry basket, in a classroom. The students had to enter the room and look around for one minute.

Pedro recorded the number of typical objects and unusual objects students could remember.

The results of Pedro's investigation are shown in **Table 1**.

	Number of typical objects recalled (out of 10)	Number of unusual objects recalled (out of 10)
Student A	8	3
Student B	10	5
Student C	9	5
Student D	5	2

Table 1

(a) Calculate the percentage of students who recalled **more than five typical objects**, out of all the students in the investigation.

(1)

Percentage .....

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P 7 5 5 2 4 A 0 6 4 8

9. Question 4 · part a · 1 mark

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Further practice · Attempt part a only.

(b) Calculate how many students recalled **less than five unusual objects** as a fraction of all students.

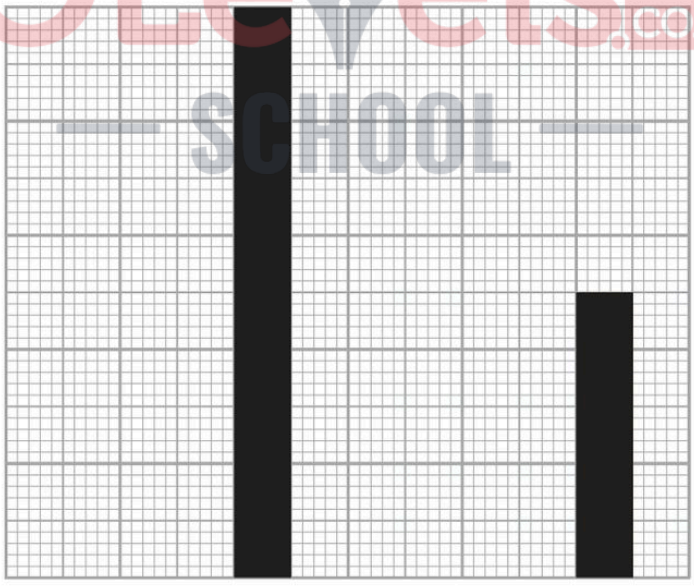
(1)

Fraction .....

(c) Complete the bar chart using the data in **Table 1**, including the label and scale for the y-axis.

(3)

A bar chart to show the number of typical objects recalled by the students



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9. Question 4 · part a · 1 mark · Working and answer

Working and answer

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